

CDS INDICES

Global Credit Derivative and Index Research

Pavan Talreja, CFA

(44-20) 3493-6802

pavan.talreja@jpmchase.com

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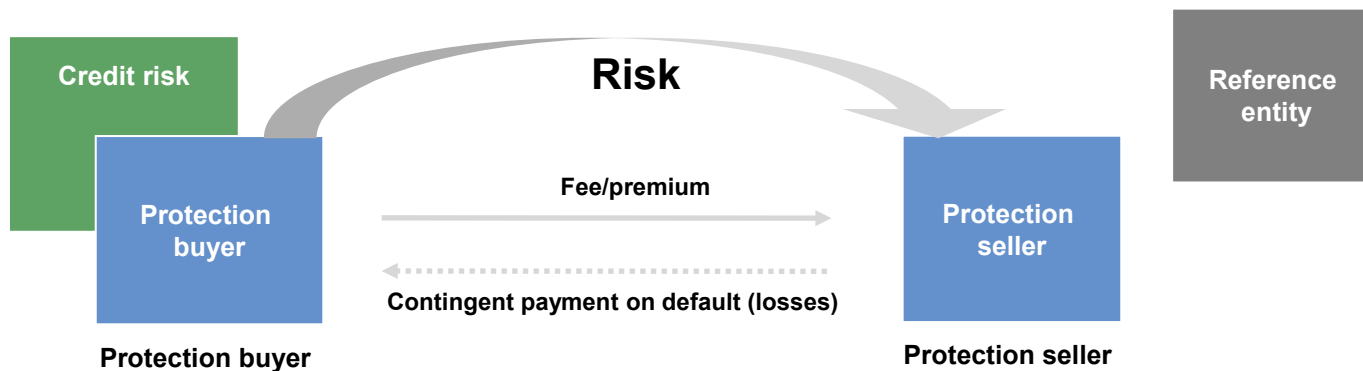
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Credit Default Swap Mechanics Recap

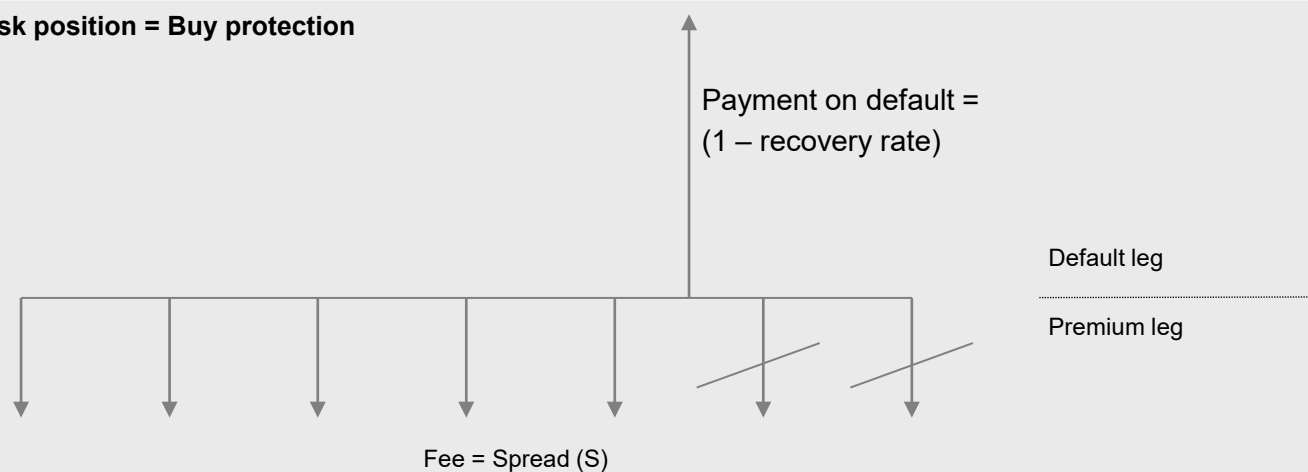
Short risk/buy protection

Long risk/sell protection



In cash flow terms (buyer of protection)

Example of a short risk position = Buy protection



Source: J.P. Morgan

Credit Default Swap Mechanics Recap

Credit Default Swaps

- There are two counterparties in a CDS contract:
 - Buyer of protection
 - Seller of protection
- The protection buyer pays the protection seller regular fees – known as the **spread**.
- In return, if the underlying company defaults, the protection seller will make a payment to the protection buyer equal to $(1 - \text{Bond Recovery}) \times \text{Notional}$.
- Each CDS contract specifies (among other things):
 - The underlying company that insurance is being bought on – the “reference entity”
 - The seniority level (usually senior unsecured debt)
 - The maturity of the contract – norm is 5 years
 - Notional of the trade
 - The contract spread/price – this quoted figure determines “fixed coupon + upfront” cash flows
 - How we define a default

Source: J.P. Morgan

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What is a CDS Index?

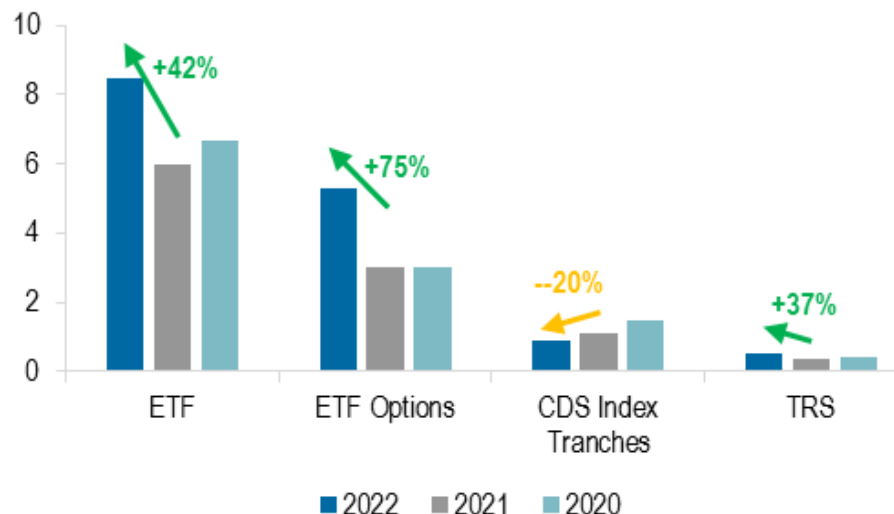
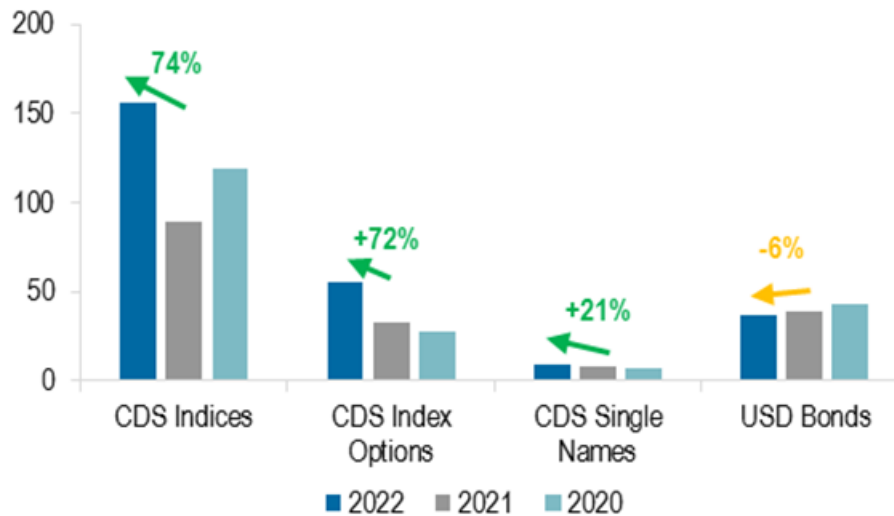
CDS Indices

- CDS indices are effectively **baskets of single name CDS**.
- e.g. iTraxx Main is a basket of 125 single name CDS contracts in European Investment Grade.
- CDS indices **trade as a single contract**, meaning that you can get diversified macro exposure to credit without trading a large number of single name CDS contracts.
- CDS indices are usually **equally weighted**: e.g. each name in iTraxx Main makes up 1/125th or 0.8% of the index notional.
- The cost of buying protection on iTraxx Main will be very similar to the cost of buying protection on all 125 single names, but there can be some differences.
- Why trade CDS indices instead of just the single names?
 - Much easier to trade one single contract than 125 different single name contracts.
 - Lower bid-offer for index.
 - Higher volumes.
 - Quick way to gain exposure to the credit risk of a region or sector.

Source: J.P. Morgan

CDS Indices are the most liquid instruments in the credit market

Average Daily Volumes in '22 vs '20-21 (\$bn)



Source: J.P. Morgan, DTCC

- Headline Index Numbers continue to grow strongly in 2022 even outstripping the volumes seen in 2020
- Single name CDS volumes grow but at a much slower pace as macro hedge demand remains very strong.

- Credit ETF Markets continue to attract more demand with rapid growth in ETF Options in particular.
- Tranche markets remain more subdued while TRS reverses the move lower in volumes in 2021.

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Global CDS Indices

European CDS Indices: iTraxx Europe Family

iTraxx Main (Investment Grade)

+ Senior / Sub Financials sub-indices

+ Main ESG

iTraxx Crossover (High yield)

Asia-Pacific CDS Indices:

iTraxx Japan

iTraxx Asia Ex-Japan IG

iTraxx Australia

US CDS Indices: CDX Family

CDX.NA.IG

CDX.NA.HY

Emerging Markets CDS

CDX.EM

Legacy indices:

- iTraxx SovX Western Europe
- iTraxx SovX CEEMEA
- iTraxx Corp CEEMEA
- iTraxx HIVOL
- iTraxx Asia Ex-Japan HY
- CDX Latin America

Source: J.P. Morgan

Bloomberg screen for CDS Indices: {CDX <GO>}

97) Settings ▾							CDS Index Monitor						
							Historical Data		Spread ▾		3 Months ▾		
Markit Indices		Spread	Chg	Spread	Basis	Roll	Low	Range	High	Avg	+/-	3M Chg	
1) Americas							◆ Avg ● Now						
10) CDX Investment Grade	⌵	64.56	-0.50		-23.6	2.4	64.6	●—◆	82.2	71.7	-7.1	-13.5	
11) CDX High Yield	⌵	105.80*	+0.08	—	3.0	0.2	98.6	—●—	105.8	102.2	+3.6	+7.7	
12) MCDX	⌵	166.53	-0.30	—		16.4	166.5	●—◆	257.8	205.3	-38.7	-91.3	
13) CDX Latin America	⌵	156.09	-0.07	—			153.5	—●—	169.8	161.7	-5.6	+2.6	
2) EMEA													
20) iTraxx Europe	⌵	53.08	-1.26	—	-2.6	3.4	52.6	●—◆	72.4	61.4	-8.4	-19.9	
21) iTraxx Crossover	⌵	313.57	-10.57	—	-21.4	16.9	313.6	●—◆	415.2	364.5	-50.9	-119.7	
22) iTraxx Sr Financial	⌵	59.39	-1.90	—	1.2	8.6	59.4	●—◆	86.1	72.1	-12.7	-27.2	
23) iTraxx Sub Financial	⌵	126.59	-2.47	—	7.9	15.3	126.6	●—◆	184.6	153.4	-26.8	-59.6	
24) iTraxx SOVX CEEMEA	⌵												
3) Asia													
30) iTraxx Japan	⌵	60.71	-0.09	—	5.3	9.9	57.4	—●—	69.7	60.9	-0.2	-9.4	
31) iTraxx Asia Ex Japan IG	⌵	65.03	+1.06	—	-5.0		64.5	●—◆	101.9	80.6	-15.6	-37.7	
32) iTraxx Australia	⌵	63.28	-1.25	—	-10.9		63.3	●—◆	102.4	85.1	-21.8	-39.7	
Emerging Markets													
40) CDX Emerging Market	⌵	96.78*	+0.06	—	-1.5	-0.9	91.7	—●—	96.8	95.5	+1.3	+5.4	

European CDS Indices

iTraxx Main

- 125 European **Investment Grade** Companies
- Equally weighted
- Sector constraints (sector buckets filled based on constituent liquidity within min-max sector limits):
 - 24-36 Autos & Industrials
 - 20-30 Consumers
 - 16-24 Energy
 - 16-24 TMT
 - 30 Financials
- Constituents must be rated higher than BBB-/Baa3 Neg Outlook – stricter definition vs cash indices.
- Composition is based on the names with the highest trading volumes.
- 1y, 3y, 5y, 7y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + **100bp** fixed coupon.
- €65bn daily volume across entire market.

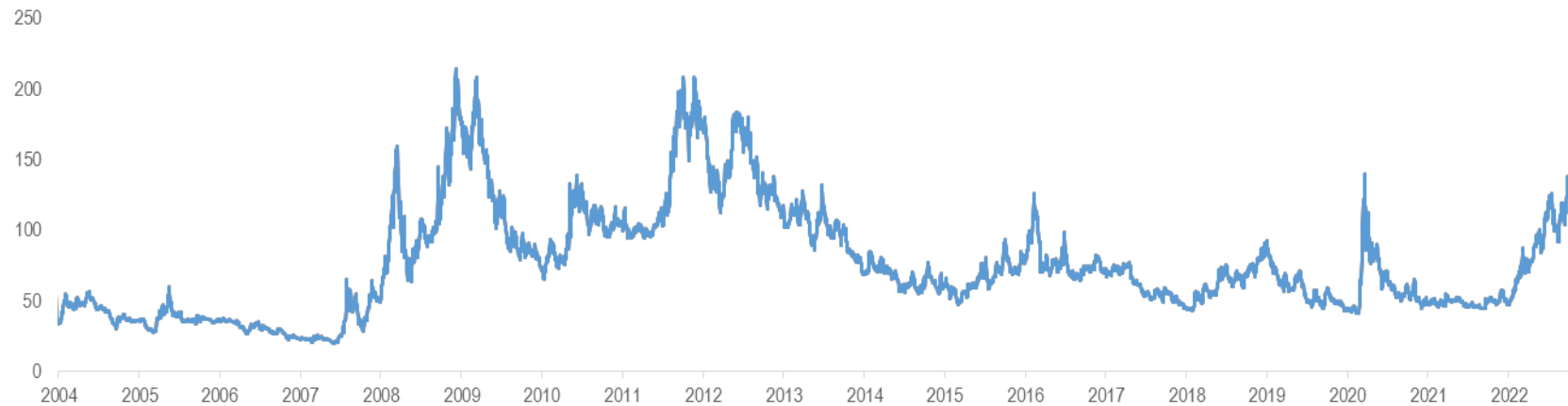
Source: J.P. Morgan

iTraxx Crossover

- 75 European **High Yield** Companies
- Equally weighted
- Financials not allowed (except Specialty Finance)
- No sector constraints
- Constituents must be rated BBB-/Baa3 Neg Outlook or below.
- Unrated issuers are allowed as long as CDS spread is greater than 1.5 x spread of Main Non-Financials.
- No distressed – 500bp upfront < 50pt
- New CDS contracts can be introduced via Suppl. List.
- Composition is based on the names with the highest trading volumes.
- 3y, 5y, 7y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + **500bp** fixed coupon.
- €14bn daily volume across entire market.

iTraxx Main and Crossover Historical Spreads

iTraxx Main 5y Spread



iTraxx Crossover 5y Spread



Source: J.P. Morgan

European CDS Indices - continued

iTraxx Senior Financials

- 30 European Banks and Insurance Companies
- Equally weighted
- Subset of 30 Financials that appear in iTraxx Main.
- **References the Senior Unsecured debt** (SLAC where available otherwise Senior Preferred/HoldCo).
- 5y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- €4.5bn daily volume across entire market.

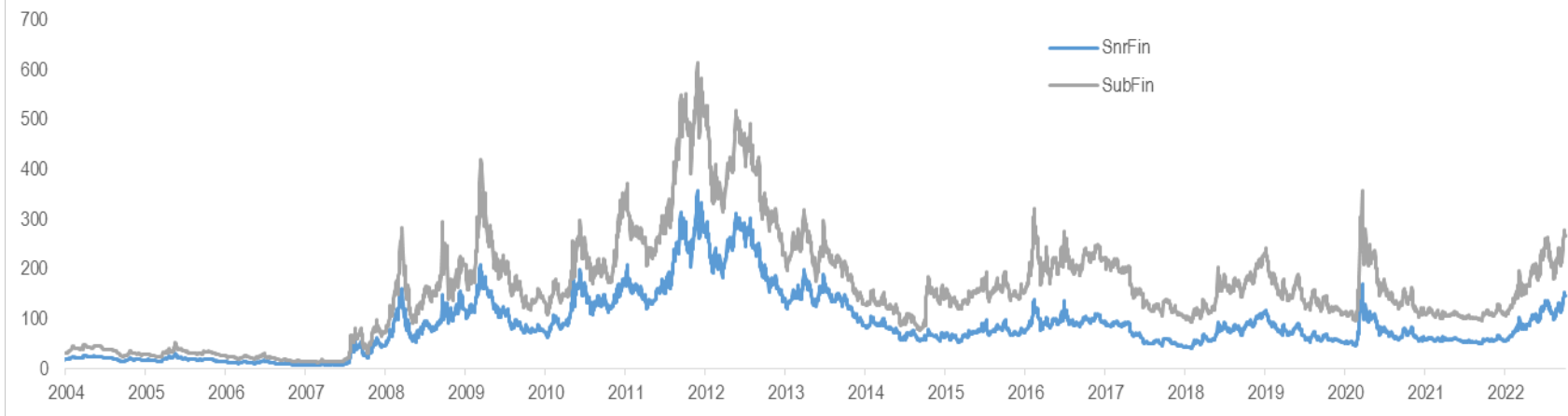
iTraxx Subordinated Financials

- 30 European Banks and Insurance Companies
- Equally weighted
- Same 30 names as Senior Financials.
- **References the Lower Tier 2 debt.**
- 5y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- €700m daily volume across entire market.

Source: J.P. Morgan

iTraxx Senior and Subordinated Financials Spreads

iTraxx Senior and Sub Spreads



Sub/Senior Ratio



Source: J.P. Morgan

US CDS Indices

CDX Investment Grade

- 125 North American Investment Grade Companies
- Equally weighted
- Very few banks and financials
- Constituents must be rated higher than BBB-/Baa3 Neg Outlook.
- Composition is based on the names with the highest trading volumes.
- 1y, 3y, 5y, 7y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- \$62bn daily volume across entire market.

Source: J.P. Morgan

CDX High Yield

- 100 North American High Yield Companies
- Equally weighted
- Financials not allowed
- Constituents must be rated BBB-/Baa3 Neg Outlook or below.
- Composition is based on the names with the highest trading volumes.
- Additional mechanism to include new names to increase weightings of unrepresented sectors
- 3y, 5y, 7y and 10y points are traded (5y most liquid).
- **Quoted on price**
 - Price = \$100 – Upfront
 - e.g. price of \$99 means buyer pays 1% upfront.
- Trades on upfront + 500bp fixed coupon.
- \$16bn daily volume across entire market.

CDX IG and CDX HY Spreads

CDX IG Spreads



CDX HY Spreads



Source: J.P. Morgan

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CDS Index Mechanics

CDS Index Mechanics

- CDS indices are quoted and traded in the same way as single name CDS.
- Quoted on spread basis, which is used to calculate traded upfront amount.
- Example:
 - We buy iTraxx Main protection on €100million for a 5y maturity.
 - Quoted spread is 75bp
 - We will pay the fixed coupon of 100bp.
 - Due to the difference in the quoted spread and fixed coupon we will also receive an upfront amount, equal to 1.125 % of notional.
- CDX HY is quoted on price, not spread. $\text{Price} = 100 - \text{Upfront}$.
- The ISDA Standard Model is used to calculate upfronts from spreads.
- You can download the ISDA Standard Model at www.cdsmodel.com (also implemented in CDSW menu on BBG).

Source: J.P. Morgan

CDSW calculator on BBG is the industry standard for mark-to-market calculations



Source: Bloomberg

Mark-to-market (clean)

CDS Index Series and Rolls

CDS Index Series

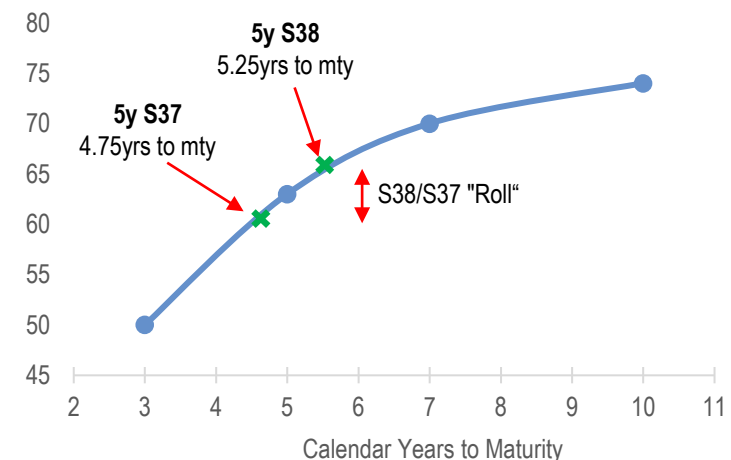
- CDS indices trade in what are known as “Series”.
- A new series is launched every six months on **20th March** and **20th September** – this is known as the “index roll”.
- The new series of each index has a **new composition** and **extra six months of maturity**.
- The latest series is known as the “**on-the-run**” index.
- In September 2022 the “on-the-run” series was Series 38 for iTraxx indices and Series 39 for CDX indices.
- For example, on 20th September 2022 we rolled from iTraxx S37 to iTraxx S38.
 - iTraxx S38 indices will have an updated composition reflecting liquidity and rating changes over the previous 6 months
 - iTraxx S38 5y maturity is 20-Dec-2027, 6 months longer than the iTraxx S37 5y maturity of 20-Jun-2027.
- Old indices become “off-the-run”. They still trade but become less liquid over time.

Rolls in Practice

- Switching a position from an old series to a new series is known as “rolling” the position.
- For example, an investor who had bought protection on iTraxx Main S37 could roll to the new series by selling back protection on Main S37 and buying protection on Main S38.
- Rolls are optional; if investors do nothing then they continue to hold the position in the old series.
- The majority of positions held by investors are rolled to the new series in order to maintain liquidity.

Source: J.P. Morgan

Roll Illustration (maturity extension only)



J.P.Morgan

JPMorgan screen for CDS Indices: {JPTE <GO>}

CDS Index						
J.P.Morgan CDS Index						
JP Morgan Electronic Execution -> Credit Derivatives -> CDS Global Index -> CDS Index -> CDS Index				MSG Contributor	10:48:23	
CDS Indices are owned by Markit				Zoom	100%	
GLOBAL CDS INDICES						
ITRAXX 5YR	BID	ASK	CHANGE	BID SIZE(MM)	ASK SIZE(MM)	Time
1) EUROPE S35	50.3600	50.6900	0.03	150	150	10:42
2) XOVER S35	244.4500	245.9500	0.70	35	35	10:43
3) FIN SNR S35	57.4600	57.9400	0.20	100	100	10:33
4) FIN SUB S35	104.9500	106.3500	-0.10	25	25	10:33
5) ESG S35	45.2750	48.2750	0.03	50	50	10:42
APAC ITRAXX 5YR						
6) AXJ IG S35	69.8500	70.6500	1.00	10	10	07:31
7) AUS S35	62.0000	62.5000	0.00	10	10	07:31
8) AXJ IG S35 Rst			0.00			
CDX 5YR						
9) NA IG S36			-0.75			04/08
10) NA HY S36			0.12			04/08
11) EM S35			0.19			04/08

CDS Index						
J.P.Morgan Index Roll Trading						
JP Morgan Electronic Execution -> Credit Derivatives -> CDS Global Index -> CDS Index -> Index Roll Trading				MSG Contributor	10:48:55	
CDS Indices are owned by Markit				Zoom	100%	
GLOBAL CDS INDICES						
ITRAXX 5YR	BID	ASK	CHANGE	BID SIZE(MM)	ASK SIZE(MM)	Time
1) MKIT Roll ITRX EUR S34/35 5YR	8.1900	8.2300	0.14	500	500	10:45
2) MKIT Roll ITRX XOV S34/35 5YR	31.2500	31.4200	0.05	200	200	10:43
3) MKIT Roll ITRX SNF S34/35 5YR	6.7900	6.9100	0.08	250	250	10:48
4) MKIT Roll ITRX SBF S34/35 5YR	13.3500	13.6500	0.17	100	100	10:48
5) MARKIT ITRX Roll ESG S34/35 5Y	6.6250	9.6250	0.00	50	50	10:42
APAC ITRAXX 5YR						
6) MKIT Roll ITRX AUS S34/35 5YR	6.2000	6.4000	0.00	10	10	07:31
7) MKIT Roll ITRX XJP S34/35 5YR	7.6000	7.8000	-0.05	10	10	07:52
CDX 5Y						
8) MARKIT Roll CDX IG S35/36 5YR			-0.01			04/08
9) MARKIT Roll CDX HY S35/36 5YR			-0.02			04/08
10) MARKIT Roll CDX EM S34/35 5YR			0.00			04/08

Source: Bloomberg

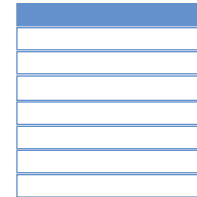
CDS Indices and Credit Events

Credit Events

- CDS indices handle credit events in the exact same way that a basket of single name CDS.
- When a credit event happens the single name splits out from the CDS index and is settled separately like a normal single name CDS contract.
- The defaulted single name is not replaced; the CDS index continues to trade on a reduced notional.
- Example:
 - We buy €125mm of iTraxx Main protection (125 names), equivalent to €1mm protection on each of the single names.
 - There is a credit event on one name with a 40% recovery.
 - We receive €600k = €1mm x (1-40%) from the single name portion of the index.
 - The index will continue to trade and will now be known as “Version 2” of the Series. The new reduced notional of the index will be €124mm.

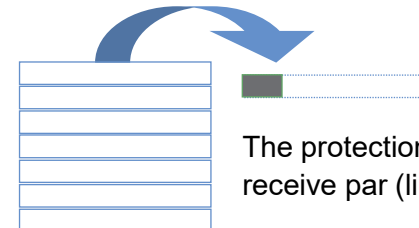
Source: J.P. Morgan

1 Credit Event on one of the names



The protection buyer and the protection seller can trigger the contract

2 The triggered name is stripped out of the contract and settled separately like a normal single name CDS.



The protection buyer will deliver the bonds and receive par (like a normal CDS)

3 The contract remains live on a reduced notional



Premium is paid or received on a reduced notional

Example Relative Value Trades with CDS Indices

- CDS indices allow investors to take relative value views across different dimensions: time, portfolios, and markets
- The four most popular relative value trades: index basis, curve, roll, and cross-index
- **Index basis trades:** trading the index vs. the underlying single-name CDS in the portfolio
 - Example: Buy Main Series 38 5y protection and sell 5y protection on the individual 125 single-name CDS in the Series 38 basket
- **Cross-index trades:** relative value between different markets
 - Example: CDX.IG vs S&P 500, Main vs VSTOXX, Main vs Xover or Main vs CDX.IG, etc
- **Curve trades:** relative value across the tenors of a single index
 - Example: 5s10s CDX.IG on-the-run duration-weighted steepener
- **Roll trades:** relative value across different Series of the same index family
 - Example: Sell protection Xover Series 37 5y and buy protection Xover Series 38 5y

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Traded vs. Theoretical Spread: Basis to Theoretical

Basis to theoretical

- The traded spread of a CDS index is driven by demand and supply.
- We can also calculate a theoretical spread for the index using the spreads of the underlying single names.
- These spreads will often not be the same – the difference is known as the basis-to-theoretical (aka index skew).
- **Basis to theoretical = Traded Index spread – Theoretical Spread**
- Why does the basis exist?
 - Both idiosyncratic factors and macro factors can be the driving force in market moves.
 - In environments where macro factors are more important, investors are more likely to express views using the index than single names. The index is likely to go wider first during sell-offs (causing the basis-to-theoretical to become more positive) and go tighter first during rallies (basis becomes more negative).
 - In idiosyncratic default risk-led environments single-name hedging drives markets. In this case single name CDS moves wider first during sell-offs.
- If the basis gets too large, investors will look to arbitrage the difference.

Source: J.P. Morgan

Trading the Basis

Basis to theoretical

- The index and underlying single names hold exactly the same default risk.
- If the basis-to-theoretical is very positive or negative, investors will look to arbitrage the difference by trading both the index and all the underlying single names.
- Example:
 - iTraxx Main is trading at a spread of 100bp and the underlying single names give a theoretical spread of 90bp.
 - Basis to theoretical = +10bp.
 - We sell protection on iTraxx Main at 100bp and buy protection on all 125 single names.
- To make a trade worthwhile the basis needs to be large enough to offset the cost of the bid-offer on the index and single names.
- These trades are known as *basis-to-theoretical trades* or *arb lists*.
- Arb lists in general are difficult to execute, given the problems associated with trying to line up a large number of single name trades simultaneously.

Source: J.P. Morgan

Calculating the Theoretical Index Spread

Theoretical Spread

- We can calculate a theoretical index spread using single name trading levels.
- As a starting point, the upfront of the index should be equal to the sum of the upfronts of all the single names, adjusted for their weighting within the index.

$$U_I = \sum_{i=1}^N \omega_i U_i$$

U_I = Index Upfront
 U_i = Single Name Upfront
 ω_i = Weighting of name in index
 N = Number of names in index

- If the index is equally weighted (as most are), then $\omega_i = 1/N$. We can also express the upfronts in terms of risky annuities, spreads and coupons.

$$RA_I (S_I - C) = \frac{1}{N} \sum_{i=1}^N RA_i (S_i - C)$$

S_I = Index Spread
 S_i = Single Name Spread
 RA_I = Index Risky Annuity
 RA_i = Single Name Risky Annuity
 C = Fixed Coupon

- We can rewrite the index risky annuity RA_I as the sum of the individual single name risky annuities $RA_I = \frac{1}{N} \sum_{i=1}^N RA_i$

- This gives us:
$$\frac{S_I}{N} \sum_{i=1}^N RA_i - \frac{C}{N} \sum_{i=1}^N RA_i = \frac{1}{N} \sum_{i=1}^N RA_i S_i - \frac{C}{N} \sum_{i=1}^N RA_i$$

- The coupons cancel out and we can then express S_I as
$$S_I = \frac{\sum_{i=1}^N RA_i S_i}{\sum_{i=1}^N RA_i}$$

- **Wider names have lower risky annuities and therefore a lower weighting** – they are more likely to default so we expect to pay the spread for a shorter amount of time.
- The lower weighting of wider spreads means that the Theoretical Spread is lower than a simple average of the single name spreads.

Source: J.P. Morgan

Historical Basis-to-Theoretical



Source: J.P. Morgan

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Additional Material

Additional Material

- [Credit Derivatives Handbook](#), 2006, J.P. Morgan
- [Global CDS Indices](#), 2021, J.P. Morgan
- [Chesapeake Corp Effect on Credit Derivatives](#), 2020, J.P. Morgan
- [Matalan Impact on Credit Derivatives](#), 2020, J.P. Morgan
- Rules and compositions available from <https://www.markit.com/Documentation/>
- [CD Player - Mid-Year Outlook 2022](#), 2022, J.P. Morgan

Source: J.P. Morgan

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